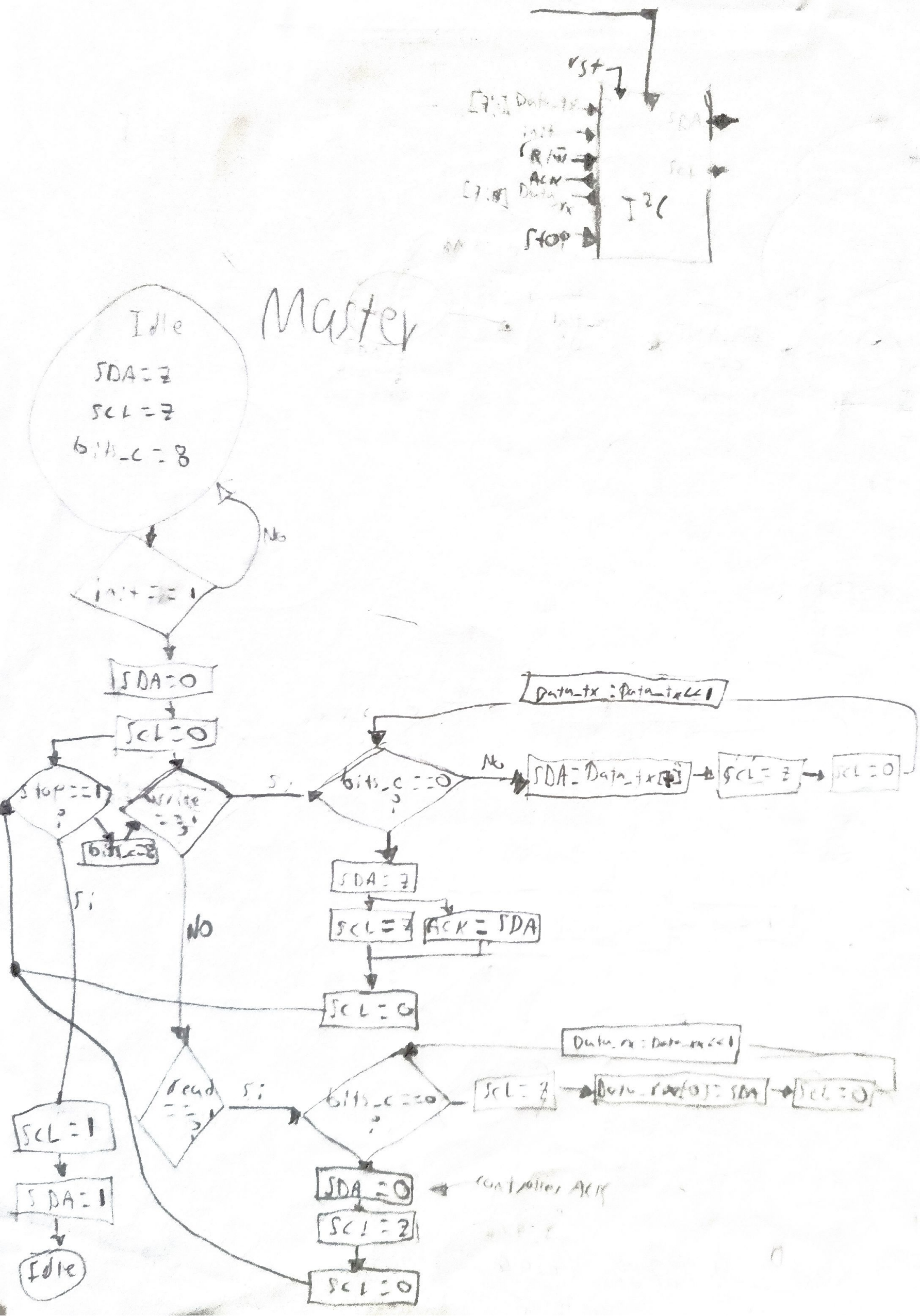
rst
[7:0] Data_tx
init
R/W
ACK
[7:0] Data_rx
STOP
SDA
SCL
I²C
Master
Idle
SDA = 1
SCL = 1
bits_c = 8
init == 1
No
SDA = 0
SCL = 0
stop == 1 ?
write == 1 ?
Yes
bits_c = 8
Yes
bits_c == 0 ?
No
SDA = Data_tx[7]
SCL = 1
SCL = 0
Data_tx = Data_tx << 1
SDA = 1
SCL = 1
ACK = SDA
SCL = 0
NO
read == 1 ?
Yes
bits_c == 0 ?
SCL = 1
Data_rx[0] = SDA
SCL = 0
Data_rx = Data_rx << 1
SDA = 0
controller ACK
SCL = 1
SCL = 0
SCL = 1
SDA = 1
Idle